

Abstract No.

Multiomic landscape of brain invasion in pediatric gliomatosis cerebri

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Pediatric diffuse gliomas are a group of devastating childhood brain cancers that span diverse histomolecular subtypes. Gliomatosis cerebri (GC) represents a radiologically defined, highly invasive growth pattern that can occur across distinct glioma subtypes. The molecular and cellular basis of the GC phenotype remains largely elusive, partly because limited clinical annotation has constrained access to tumor material for systematic investigation. To address this, we established a multinational GC consortium assembling tumor specimens from 21 primary pediatric GC patients across the US and Europe (1–18 years). We developed an integrated multi-center framework combining clinical, radiologic, histomolecular, and multiomic tumor profiling, including single-nucleus RNA sequencing (snRNA-seq; Smart-seq2), spatial transcriptomics (10x Xenium), and spatial proteomics (imaging mass cytometry), with complementary patient-derived functional modeling. Consistent with previous reports, pediatric GC predominantly arises in H3/IDH-wildtype pediatric high-grade glioma (pedHGG; 17/21), particularly the methylation-defined subclass pedHGG_RTK2A (7/21). Genomic profiling reveals recurrent alterations in *EGFR* (7/21), *PIK3CA/PIK3R1* (7/21 combined), *TERT* (7/21), and *BCOR* (6/21). Our snRNA-seq analyses reveal marked inter- and intratumoral heterogeneity, with recurrent AC-like, OPC-like, NPC-like, and MES-like malignant cell states identified across the cohort, suggesting an interplay of multiple cellular programs underlying GC invasion. Building on these findings, our integrated spatial multiomic analyses define the *in situ* organization of cell populations and assess invasion-permissive cellular niches. Complementary patient-derived tumor–neuronal co-culture and xenograft models further interrogate how tumor-intrinsic cellular programs and microenvironmental interactions cooperate to drive brain-wide dissemination in pediatric GC. Together, this work establishes the first comprehensive, high-resolution atlas of pediatric GC and provides an integrated view of the molecular, cellular, and spatial features associated with diffuse brain invasion. These insights may enable more precise biological classification and inform new avenues for therapeutic intervention in this dismal pediatric disease.



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